

# 2024 Bi H1 Q23

## Section: Sustainability and Interdependence

### Topic: Symbiosis

#### Question Summary:

Three examples describe interactions between species. The question asks which of them are mutualistic relationships.

#### Worked Solution:

- Statement 1: Ants feed on sugary liquid (honeydew) produced by aphids and protect the aphids from predators. Both species benefit -> mutualism.
- Statement 2: Coral and zooxanthellae exchange nutrients — the coral provides protection and CO<sub>2</sub>, while the algae supply sugars -> mutualism.
- Statement 3: Cleaner wrasse remove parasites from larger fish (e.g., sharks) and gain food in return -> mutualism.

Since all three relationships involve mutual benefit, all are mutualistic.

Final Answer: **D** (1, 2, and 3 — all are mutualistic).

#### Revision Tips:

- Mutualism: both species benefit (e.g., pollinators and flowers, cleaner fish).
- Parasitism: one benefits, the other is harmed (e.g., tapeworms).

- Commensalism: one benefits, the other is unaffected.